

Anika Tasnim Ritu

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EDUCATION

Master of Science in Electrical & Electronic Engineering April 2023 – Jan 2026
Rajshahi University of Engineering & Technology (RUET), Rajshahi-6204, Bangladesh.
CGPA: **4.00/4.00**.

Bachelor of Science in Electrical & Electronic Engineering Feb 2017 – Nov 2022
Rajshahi University of Engineering & Technology (RUET), Rajshahi-6204, Bangladesh.
CGPA: **3.90/4.00** (Ranked 3rd out of 123 students).

RESEARCH INTERESTS

Robust and Interpretable Healthcare AI; Biomedical Signal Processing; Wearable Physiological Sensing; Trustworthy Human Sensing; Self-Supervised and Multimodal Healthcare AI; Medical Image Analysis.

RESEARCH EXPERIENCE

Remote Research Assistant Apr 2025 – Present
Health Informatics Research Laboratory (HIRL) ([Lab Website](#)), Department of Computer Science and Engineering (CSE), Daffodil International University (DIU), Dhaka, Bangladesh.

Research Supervisor: Dr. Md. Zahid Hasan, Associate Professor, Dept. of CSE, DIU.

Undergraduate and Postgraduate Researcher Jan 2021 – Jan 2026
Department of Electrical and Electronic Engineering, Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh.

Research Advisor: Dr. Ajay Krishno Sarkar, Professor, Dept. of EEE, RUET.

SELECTED PUBLICATIONS

Journal Articles

1. **A. T. Ritu**, S. Hossain, M. Z. Hasan, S. Bosak, A. Hanip, and T. Bhuiyan, "Interpretable CNN–LSTM Framework for Multiclass Neuromuscular Disorder Classification Using Clinically Relevant EMG Features," **IEEE Transactions on Artificial Intelligence**, vol. 7, no. 6, pp. 3383–3398, 2026. [[Link](#)]
2. S. Bosak, M. R. Islam, M. R. I. Sheikh, **A. T. Ritu**, M. B. Hossain, and M. J. Hossain, "Robust Control of Renewable-Hydrogen Electrolyzer-Buck Converter Systems Using Modified Double Integral Sliding Mode Controller for Grid Ancillary Services," *Journal of Power Sources*, vol. 677, p. 239953, 2026. [[Link](#)]

Journal Manuscripts Under Review

1. M. Z. Hasan, S. Hossain, S. M. D. Alam, S. K. Shehjad, A. Dey, M. Tazakkaruzzaman, **A. T. Ritu**, M. Hasan, and A. Moustafa, "Domain-Adaptive Deep Learning for Robust Multi-Class Retinal Disease Screening Across Clinical Environments," submitted to *Discover Artificial Intelligence*.
2. S. Hossain, M. M. Sayeb, T. A. Patwary, M. Mehenaj, **A. T. Ritu**, M. M. Rahman, M. Z. Hasan, J. Mumin, S. Ahmmed, and O. Jaman, "EDNet-20: Enhancing Multi-Label Ocular Disease Detection through Deep Learning Optimization," submitted to *Discover Artificial Intelligence*. [[Preprint](#)]
3. M. Z. Hasan, R. Ibrahim, A. Fariha, N. Sakib, S. T. Jahan, M. Momtaj, M. A. Rahman, **A. T. Ritu**, S. Hossain, and A. Moustafa, "GWO-RMF: A Meta-Heuristic Residual Manifold Framework for Interpretable Neuromuscular Disorder Classification Using Clinically Relevant EMG Features," submitted to *Applied Computational Intelligence and Soft Computing* (Wiley).

Conference Papers

1. **A. T. Ritu**, A. K. Sarkar, and S. Bosak, "Statistical Analysis of Physical Activity's Impact on Cardiovascular Health using ECG and Lifestyle Metrics," in *2025 4th International Conference on Electrical, Computer and Communication Engineering (ECCE)*, pp. 1–6, 2025. [[Link](#)]
2. **A. T. Ritu**, A. K. Sarkar, and S. Bosak, "Assessing the Relationship between Bodily Biological Parameters and Cardiac States of Different Subjects," in *2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE)*, pp. 1–6, 2024. [[Link](#)]
3. H. U. Rahman, **A. T. Ritu**, H. J. Rasha, and M. S. Anower, "Comparative Analysis of Microcontrollers in Wearable Devices for Real-Time Arrhythmia Classification: An Embedded Machine Learning Approach," in *2024*

6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT), pp. 1327–1331, 2024. [\[Link\]](#)

4. M. S. Sorkar, M. M. Rahman, and **A. T. Ritu**, “Detection of Neuromuscular Disorders from EMG Signals: A Machine Learning Approach,” in *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, pp. 1–6, 2026. [\[Link\]](#)
5. M. T. Nabi, M. N. Hasan, **A. T. Ritu**, J. Jumana, and I. Z. Netto, “Touchless IoT-Based Health Monitoring System for Elderly and Maternal Healthcare Applications with Real-Time Cloud Integration,” in *International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure*, 2026. **(Accepted)**

Conference Manuscripts Under Review

1. **A. T. Ritu** and A. K. Sarkar, “Interpretable Lifestyle-Based Prediction of ECG-Derived Cardiovascular Phenotypes,” submitted to the *IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON)*, 2026.
2. **A. T. Ritu** and A. K. Sarkar, “Unsupervised ECG-Lifestyle Clustering for Early Cardiovascular Phenotyping in Young Adults,” submitted to the *IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON)*, 2026.

PROFESSIONAL EXPERIENCE

Lecturer, Department of Electrical and Electronic Engineering Jul 2023 – Present
Bangladesh Army University of Engineering & Technology, Natore, Bangladesh.

Lecturer (Contractual), Department of Electrical and Electronic Engineering Mar 2023 – May 2023
Bangladesh Army University of Engineering & Technology, Natore, Bangladesh.

TECHNICAL SKILLS

Programming: Python, MATLAB, C/C++.

ML/DL: PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas, Explainable AI (XAI).

Simulation: MATLAB/Simulink, Proteus, Multisim.

Hardware: Arduino, STM32, IoT devices and sensors.

Research Tools: L^AT_EX, Microsoft Office, Git, GitHub, Microsoft Visio, Draw.io, EndNote.

AWARDS & ACHIEVEMENTS

Government Merit-based Scholarship (4×), Ministry of Education, Bangladesh

Awarded for top performance in four of Bangladesh’s national public examinations, with scholarship tenure extending from primary education through undergraduate studies:

Higher Secondary School Certificate – Talent Pool Scholarship; Ranked **7th** in Dinajpur Board. 2017 – 2022

Secondary School Certificate – Talent Pool Scholarship; Ranked **13th** in Dinajpur Board. 2015 – 2016

Junior School Certificate – General Scholarship. 2012–2014

Primary School Certificate – Talent Pool Scholarship; Ranked **18th** in Dinajpur Board. 2009–2011

PROFESSIONAL ACTIVITIES

Publication Committee Member, IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON), 2026.

Student Member, IEEE.

REFERENCES

Dr. Ajay Krishno Sarkar

Professor

Dept. of EEE, RUET

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Dr. Md. Rafiqul Islam Sheikh

Professor

Department of EEE, RUET

Email: mri.sheikh@eee.ruet.ac.bd